Resistivity Detection of Perfluoroalkyl Substances with Fluorous Polyaniline in an Electrical Lateral Flow Sensor

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1. Motivation

Per- and polyfluoroalkyl substances (PFAS), including perfluorooctanoic acid (PFOA) and perfluorobutanoic acid (PFBA), represent one of the most persistent classes of anthropogenic contaminants, widely known as "forever chemicals" due to their carbon-fluorine backbone that resists thermal, chemical, and biological degradation [1]. Over the past several decades, PFAS have been widely used in nonstick cookware, textiles, and packing materials for their hydrophobic and oleophobic properties [2]. However, their exceptional stability has led to global distribution in surface water, groundwater, and even drinking water supplies.

Chronic PFAS exposure has been associated with serious human health effects including thyroid disease, liver damage, immunotoxicity, and several cancers [1, 3]. In response to mounting evidence of harm, the U.S. Environmental Protection Agency (EPA) recently set maximum contaminant labels (MCLs) at 4 ppt for PFOA and 1 ppt for PFBA in drinking water [4]. However, these ultratrace concentrations are challenging to detect using conventional analytical methods.

Currently, liquid chromatography coupled with mass spectrometry (LC-MS) remains the gold-standard technique for PFAS quantification. Although LC-MS achieves sub-ppt sensitivity, it requires costly instrumentation, complex sample preparation, and trained personnel, limiting accessibility to centralized laboratories [5]. Consequently, most small communities and developing regions lack the infrastructure to monitor PFAS in local water supplies.

This technological gap underscores the urgent need for low-cost, portable, and field-deployable detection platforms that can rapidly quantify PFAS on-site with high sensitivity and selectivity. Such technologies would enable proactive environmental monitoring and public-health protection allowing users to test residential water sources or contaminated sites without the need for specialized facilities.

Recent advances in electrical lateral flow assays (e-LFAs) offer promising solutions. By coupling conductive polymers with tailored surface chemistries, these sensors can transduce molecular recognition events into measurable electrical signals. The fluorous polyaniline (F-PANI) platform developed by Park et al. (2024) demonstrates the feasibility of detecting PFOA at concentrations as low as 400 ppt and PFBA at 200 ppt, highlighting the potential for rapid, inexpensive PFAS screening [6], Yet, further engineering improvements are needed to achieve faster assay times and EPA-level detection limits. Developing next-generation e-LFA

sensors for PFAS detection could therefore bridge a crucial gap between laboratory-grade analytical methods and practical field monitoring, contributing to environmental remediation efforts and safeguarding public health globally.

2. Analyte Analysis

The project focuses on perfluorooctanoic acid (PFOA), a synthetic C_8 perfluorinated carboxylic acid belonging to the broader class of per- and polyfluoroalkyl substances (PFAS). Its molecular structure consists of an eight-carbon fully fluorinated alkyl chain attached to a terminal carboxylic acid group (-COOH). The strong carbon-fluorine bonds (bond energy ≈ 485 $kJ \cdot mol^{-1}$) imparts exceptional chemical and thermal stability, rendering PFOA resistant to degradation by hydrolysis, photolysis, and microbial activity [1, 2].

PFOA is most commonly encountered in groundwater, surface water, and drinking water near industrial discharge sites, airports, and military bases, where it originates from non-stick coatings and textile treatment [2]. Concentrations vary widely by source, typical background levels in natural waters are below 10 ppt, while contaminated sites can reach the ppb range [7,8].

At environmental pH (6-8), PFOA exist almost entirely in its anionic form (-COOH-) due to its $pK_a = 2.8 - 3.5$, allowing it to interact with positively charged or polar surfaces. The perfluorinated tail imparts strong hydrophobic and lipophobic properties, leading to surface accumulation and resistance to adsorption onto conventional sorbents. These same properties also enhance fluorous-fluorous affinity with fluorinated polymer surfaces forming the basis for selective sensing F-PANI-based electrical lateral flow assays [6].

In biological systems, PFOA is highly bioaccumulative, with reported human serum half-lives of 2-4 years, and has been linked to thyroid dysfunction, immunotoxicity, liver damage, and several cancers [1, 7]. From a detection standpoint, its environmental significance begins at concentrations as low as 4 ppt, the EPA's MCL for drinking water. PFOA is often found alongside other PFAS such as perfluorobutanoic acid (PFBA), perfluorohexanoic acid (PFHxA), and the fluorinated analog octanoic acid (OA), which serve as important controls in evaluating sensor specificity. Among these, PFOA's longer chain length and higher degree of fluorination result in stronger partitioning into the fluorous phase, making it the ideal model analyte for selective PFAS detection and environmental monitoring [6].

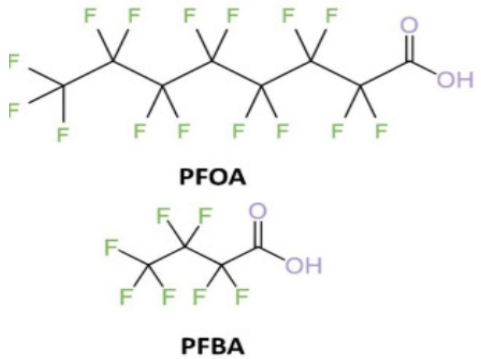

Figure 1. *Chemical structure of perfluorooctanoic acid (PFOA) and perfluorobutanoic acid (PFBA), highlighting the fully fluorinated carbon chain responsible for their chemical persistence.*

3. Sensor Reaction and Instrumentation

The operation of the fluorous polyaniline (F-PANI) sensors is fundamentally based on the reversible acid-base doping chemistry of polyaniline (PANI), a conducting polymer whose electrical conductivity is modulated by protonation of its imine nitrogen sites [9, 10]. This protonic doping reversibly converts the polymer between its insulating emeraldine base (EB) and conductive emeraldine salt (ES) form, with the conductivity difference spanning several orders of magnitude [11]. The redox-active nature of PANI and its ability to reversibly protonate and deprotonate make it an ideal material for electrochemical and chemical sensing application. In the context of PFAS detection, the PANI polymer is functionalized with a fluorous surfactant Krytox-PEG-600-diamide (KPD), giving rise to the fluorous polyaniline composite, which enhances the selective capture of perfluorinated analytes via fluorous-fluorous interactions [6]. The sensor's functionality thus arises from the coupling of two chemical processes: (1) the acid-base equilibrium between doped and undoped PANI, and (2) the partitioning and proton donation of PFAS molecules into the F-PANI layer.

The synthesis of the sensing material proceeds through a multistep sequence of polymerization, deprotonation (dedoping), and surfactant modification. First, aniline monomers are polymerized via interfacial oxidative polymerization between an aqueous oxidant layer containing camphorsulfonic acid (HCSA, $C_{10}H_{16}O_4S$) and ammonium persulfate (APS, $(NH_4)_2S_2O_8$), and an organic phase of aniline dissolved in dichloromethane. This reaction yields

nanofibrous emeraldine salt (PANI-ES), the protonated and conductive form of the polymer. The resulting material exhibits dark-green coloration and characteristic FTIR bands at 1,576 cm^{-1} (quinoid C=N) and 1,491 cm^{-1}(benzenoid C=C), confirming partial protonation of nitrogen sites. The overall oxidative polymerization can be represented by schematically as:

$$(C_6H_5NH_2)_n + (NH_4)_2S_2O_8 + HCSA \rightarrow [-C_6H_4-NH-C_6H_4-N^+H-]_n \cdot (CSA^-)_n + H_2SO_4 + H_2O$$

To obtain the neutral emeraldine base form, the doped nanofibers are dedoped by treatment of 0.1 M of ammonium hydroxide (NH_4OH), which deprotonates the amine sites and removes the CSA^-counterions. The reaction proceeds as:

$$PANI - ES \cdot (HCSA^-) + NH_4OH \longrightarrow PANI - EB + NH_4CSA + H_2O$$

The dedoped nanofibers are then functionalized with fluorous surfactant, KPD, to form the F-PANI composite. The surfactant establishes non-covalent hydrogen binding between its amide and ether groups and the nitrogen sites of PANI, while its perfluorinated tails impart strong fluorous character. This modification both prevents nanofiber aggregation and provides selective affinity for PFAS via fluorous-fluorous interactions [6]. The F-PANI dispersion is printed onto nitrocellulose (NC) membranes to create a 0.7 cm-wide test line, after drying the film forms a porous, hydrophobic, nanostructured sensing region.

During operation, a water sample containing perfluoroalkyl acids migrates along the strip by capillary action and reaches the F-PANI test line. At this stage, two coupled processes occur: (1) partitioning of PFAS molecules into the fluorous layer and (2) acidic protonation of PANI. The perfluoroalkyl carboxylic acid group (R-COOH) donate protons to the imine (-N=) sites of the emeraldine base, yielding the emeraldine salt. The overall acid-base doping reaction can be expressed as and can be seen in Figure 2:

$$[-NH-C_6H_4-N = -C_6H_4-]_n + 2R-COOH \rightleftharpoons [-NH-C_6H_4-NH^+-C_6H_4-]_n \cdot (R-COO^-)_2.$$

Here, $RCOO^-$represents the PFAS anion (e.g., $PFOA^-$ or $PFBA^-$), which serves as the counterion stabilizing the positive charge on the protonated $-NH^+-$ sites. The stoichiometry of this redox acid-base pair is 1:2, where one repeat unit of PANI-EB (containing two aniline subunits) accepts two protons from the two acid molecules to form the fully doped emeraldine salt. This reaction is reversible, upon exposure to a basic medium, deprotonation regenerates the undoped emeraldine base, releasing PFAS anion back into the solution.

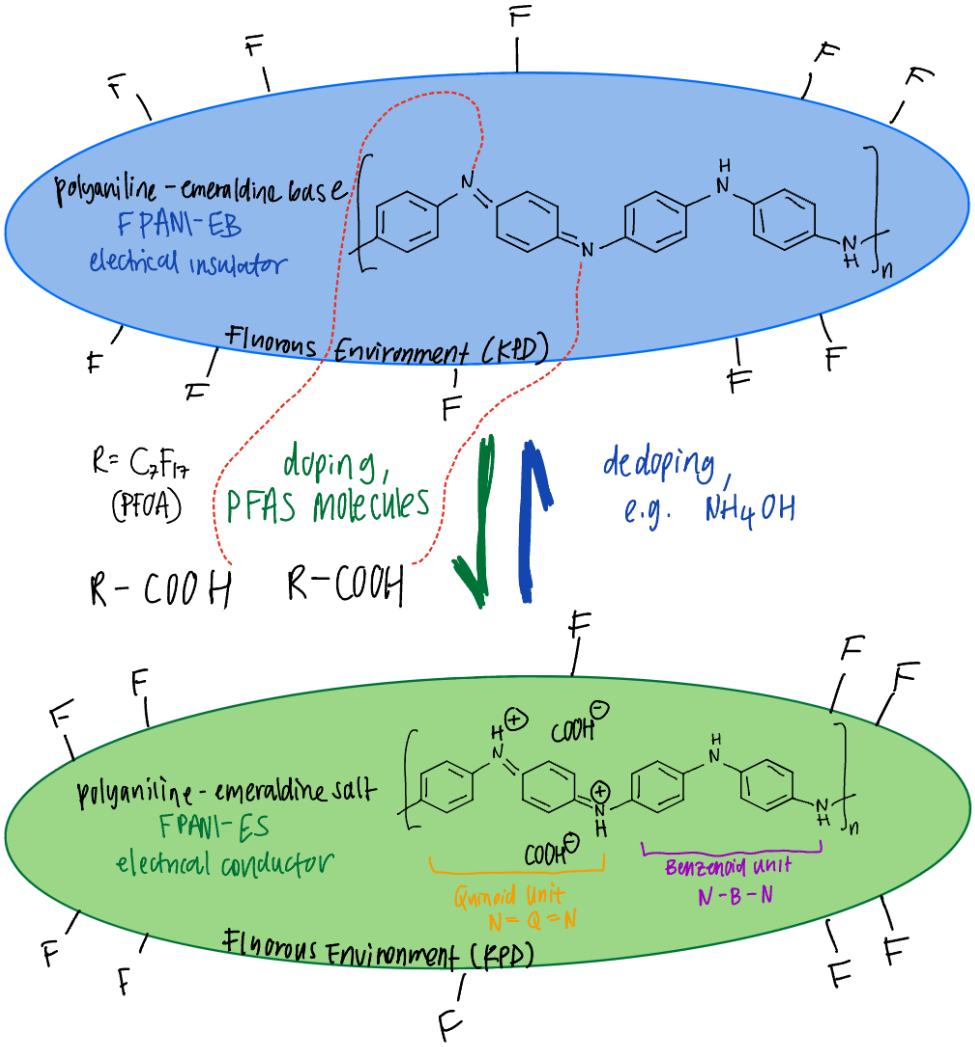

Figure 2. *Protonic doping mechanism of F-PANI by PFAS acids. Top: F-PANI in the emeraldine base (EB) state (alternating –NH– / =N–, insulating, dark blue) within a fluorous environment provided by KPD surfactant. Middle: reversible acid–base reaction; PFAS (R–COOH) donates protons to the polymer (acid doping), while base (OH^-) dedopes (reset). Bottom: emeraldine salt (ES) state with protonated –NH^+– sites stabilized by two PFAS carboxylate counterions (R–COO^-), yielding a 10^4-fold conductivity increase.*

Stoichiometry per repeat: [PANI-EB] + 2 R–COOH ⇄ [PANI-ES]·(R–COO^-)$_2$

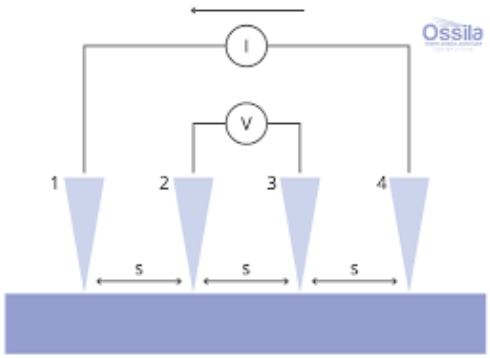

Figure 3. *Four-point probe configuration used for electrical lateral flow assays. A constant current is applied between the outer electrodes while voltage is measured between the inner electrodes to determine the film resistivity of the hydrated F-PANI test line.*

The change in conductivity is measured electrically rather than optically, significantly reducing cost and complexity. The test line resistance is measured using a four-point probe configuration connected to a Keithley 2400 source-measure unit, which applies a small current through the outer electrodes and measures voltage across inner electrodes. This setup eliminates the effect of contact resistance and allows precise measurement of film resistivity under hydration. The assay is designed so that a hydrated, doped test line exhibits sharp decrease in resistance with increasing PFAS concentration, producing a quantitative calibration curve across the range 10^{-10}-10^{-2} [6]. The sensor achieves limits of detection of 400 ppt for PFOA and 200 ppt for PFBA, corresponding to environmentally relevant concentrations near current U.S. EPA thresholds.

Unlike conventional colorimetric LFAs, this electrical readout requires only simple instrumentation:

- a nitrocellulose strip containing the F-PANI test line,
- a four-point probe for contact measurement,
- a sourcemeter (e.g., Keithley 2400) to apply low voltage and record current, and
- a portable data acquisition interface (which can be miniaturized into handheld meter).

Because the signal transduction relies on bulk conductivity rather than optical intensity, the device is less sensitive to lighting conditions or user interpretation, making it suitable for field-deployable PFAS testing in environmental and household water samples.

4. Diagram of Sensor and Process

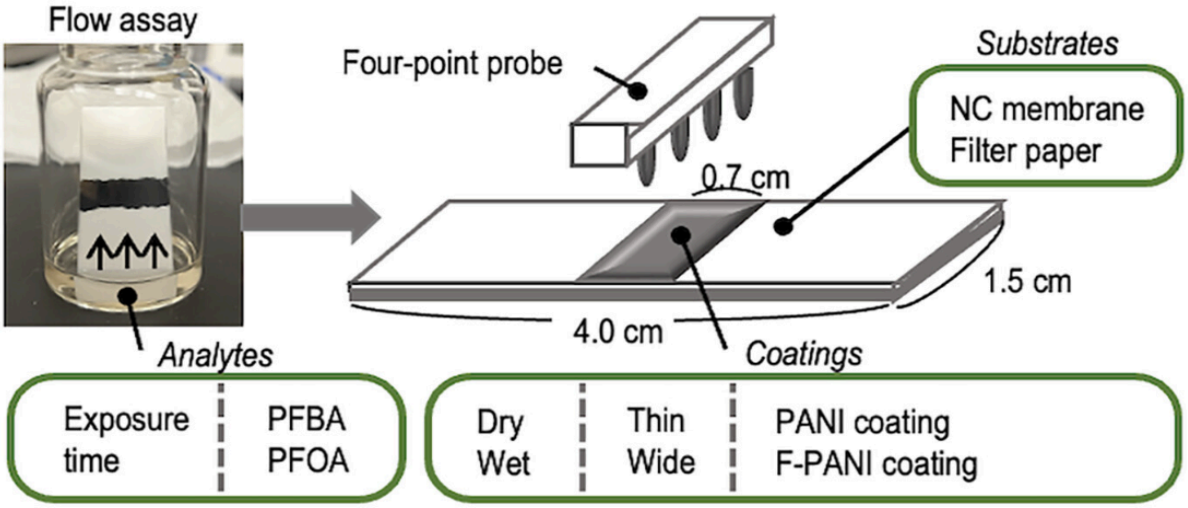

Figure 4. *Overview Diagram of Sensor Instrumentation.*

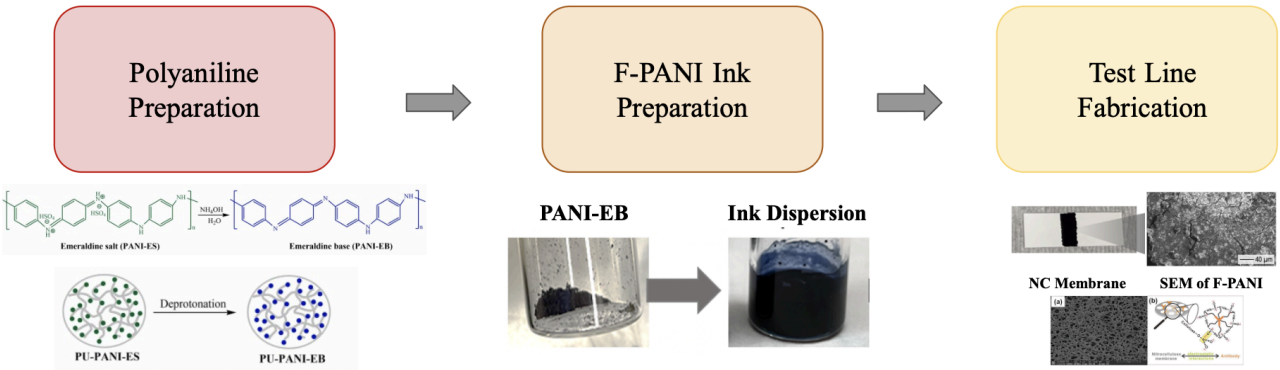

Figure 5. *Flow Diagram of the Manufacturing Process*

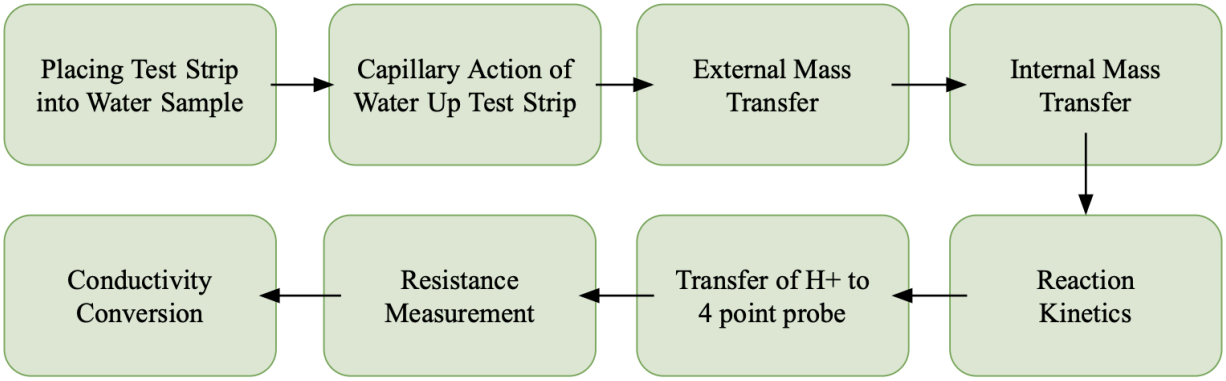

Figure 6. *Process Diagram of Test Strip.*

5. Sensitivity, Specificity, and False Results

Sensitivity of the sensor is limited to 400 ppt for PFOA and 200 ppt for PFBA are orders of magnitude higher than the US EPA set water limits. In addition, variation in test line coatings and possible microcracks may lead to variance in resistance even at a fixed concentration. The researchers found that a test line of 0.7 cm or less provided consistent results over different analyte concentrations while wider lines of greater than 0.7 cm displayed higher resistances. Variation in resistances for different test line sizes were likely due to non-uniform protonic doping through the test line. For their work, the sensor performance was determined by test line widths of 0.7 cm.

For specificity, there was a high discrimination between PFAS and non-fluorous equivalents such as butyric acid as the nonfluorous equivalent gives up to 10^4 times weaker response for the same molar concentration. From plotting the experimental results from Table 1 of Park et. al, the linear range of the sensor is determined to be from PFOA 10^{-9} M to 10^{-3} M (see Figure 7) [6].

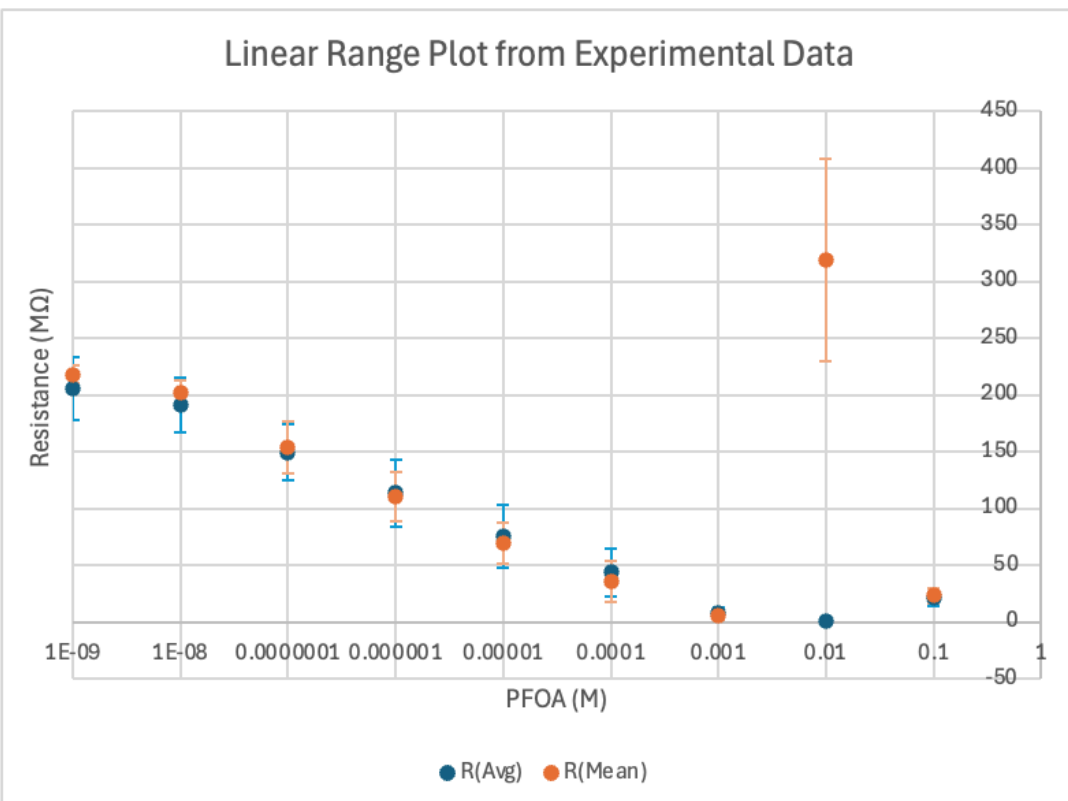

Figure 7. *Linear Range Plotted from Experimental Data from Park et. al [6].*

False positives may occur with a slightly more acidic environment where there may be some cross-reactivity or the sensor may pick up resistance signals from other molecules in the water causing the test to produce a positive result when there isn't any PFAS. To avoid this error, proper calibration of the resistance and reference pH can be taken to ensure that the water sample is around neutral pH.

False negatives of the sensor may occur when the strip is allowed to dry for 15 minutes, it may display a resistance of above 220 MΩ even if PFAS was present. To avoid this error, the researchers ensured they measured the resistance within 5 minutes after removing the strip from solution when the strip was properly hydrated. Due to the inherent resistance of water, if there is PFAS that is present at too low a concentration or other molecules that increase the resistance, that may also lead to a false negative.

6. Engineering Strategies

Several different engineering strategies were employed for the current device design. In order to maintain a high surface area for analyte interactions, PANI is polymerized into nanofibers. The process includes interfacial oxidative polymerization through camphorsulfonic acid and ammonium persulfate into PANI-ES powder. Once PANI-ES is converted into PANI-EB, the fluorous-PANI-EB ink is stabilized by noncovalent interactions between the amide and ether groups of KPD and nitrogen atoms of PANI-EB through the use of biomolecular design. The test lines on the NC membrane are designed to include a limit of detection through a spike in resistance. When the test strip is dried for 15 minutes, the water evaporates and a resistance above 220 MΩ is displayed. When exposed to an aqueous solution, the F-PANI resistance decreases. The device works in batch processes [6].

7. Cost and Time Analysis

Costs were calculated on the assumption that materials were sourced from Sigma Aldrich and associated companies since the authors of the paper sourced their materials from Sigma Aldrich [27]. The total cost for one device is $44.83, which was calculated by summing the two main steps in the process: synthesis of PANI polymer and preparation of ink and coating. In addition, capital investment was taken into account. The four point probe is estimated to cost $1399 [26]. This cost was not taken into account with the device.

Synthesis of PANI Polymer		
Name	**Amount (g)**	**Cost ($)**
Aniline	0.298016	$0.069
Dichloromethane	13.3	$6.520
Ammonium persulfate	1.584	$0.000
Camphorsulfonic acid	12.98	$0.004
NH4OH	9	$0.959
TOTAL	**37.162016**	**$7.552**

Preparation of Ink and Coating		
Name	**Amount**	**Cost ($)**
PANI-EB-powder	0.03 g	$0.006
DI Water	0.0015 L	$0.051
KPD	0.1 g	$26.000
Hexafluroisopropanol	0.002 L	$6.448
NC membrane	1 membrane	$4.770
TOTAL		**$37.275**

Total Cost per Unit		
TOTAL		**$44.827**

Capital Investment		
Four point probe		**$1,399**

Synthesis of PANI Polymer	
Solution preparation	15 minutes
Interfacial polymerization	14 hours
Filtration	15 minutes
Reduction	1 hour
Total time	**15.5 hours**
Preparation of Ink and Coating	
Prepare PANI dispersion	1 hour
Prepare surfactant solution	15 minutes
Prepare F-PANI	30 minutes
Coat membrane with ink	15 minutes
Substrate drying	2 hours
Total time	**4 hours**
Experimental	
Constant resistance	**30 minutes**
Total time	**30 minutes**

Table 1. *Technical economic analysis for creation of one device*

Table 2. *Time analysis for lateral flow assay including the synthesis of PANI polymer, preparation of ink and coating, and experimental time*

Time was calculated based on the guidelines of the paper. All squares marked with green indicate a time measurement dictated by the paper. All squares marked with gray indicate an estimated time for the step. All preparation steps were assumed to take 15 minutes. The total time for the experiment from start to finish is 20 hours.

8. Shortcomings and Areas for Improvement

1. *Transport Limitations/ Long Assay Time*

The dominant limitation in the F-PANI e-LFA design is the slow capillary transport across the hydrophobic F-PANI test line. The fluorous coating, while essential for PFAS selectivity, significantly increases the contact angle and lowers the capillary pressure that drives lateral flow, resulting in times exceeding 30 minutes before a stable resistivity signal is reached [6]. This transport delay limits the device's practicality for field or at-home testing, where a user-friendly assay typically requires readouts within 10-15 minutes, similar to commercial lateral flow tests for pathogens or biomarkers [12].

Because this time constraint arises from physical transport and interfacial diffusion processes, not from the chemical binding kinetics, our group will directly target this limitation. We will apply Washburn's capillary flow analysis and dimensionless number modeling to redesign the F-PANI test line structure. Specifically, we will explore modifications that increase pore size, reduce film thickness, and improve surface wettability, thereby accelerating transport while preserving analyte selectivity.

2. *Film Microstructure and Variability*

SEM analysis of the F-PANI coating revealed microcracks averaging ~3μm wide, produced by dehydration stresses during film formation [6]. These structural defects lead to nonuniform protonic doping across the test line, resulting in broader resistance distributions and signal noise. Microcracks can also act as unintended fluid bypass channels, reducing the effective contact time between analyte molecules and active sensing regions. While complete elimination of microcracks would require advanced polymer processing, improving the coating uniformity and hydration stability could enhance reproducibility and signal consistency.

Although important, this limitation is not the group's primary focus because film morphology optimization requires iterative materials synthesis and characterization beyond the scope of this course. However, understanding its contribution to variability will inform future recommendations.

3. *Sensitivity and Environmental Specificity*

The report limit of detection for the current F-PANI e-LFA is 400 ppt for PFAO and 200 ppt for PFBA, while impressive for a field-deployable electrical assay, remains roughly two orders of magnitude above the U.S. EPA's new drinking water MCL of 4 ppt for PFOA. Reaching this extremely low concentration range would not just require improving the existing material, it would demand a completely different sensing approach. Achieving ppt to sub-ppt detection would require extra steps to chemically concentrate the sample, or amplification methods such as enzymatic reactions or redox cycling. These changes would make the device more complex, slower, and more expensive, defeating the purpose of a simple, rapid field test.

Furthermore, the current sensing mechanism is based on direct conductivity changes in the polymer film after PFAS doping is a single step, non-amplifying process. This design makes it robust and easy to operate but limits how much the signal can be boosted by additional analyte

binding. Therefore, while improving the LOD to meet EPA standards is desirable in theory, it is not practical within the scope of this project or the intended field-usable format.

Our group chose to focus on reducing assay times rather than improving the LOD for several technical and practical reasons:

1. *Feasibility within course scope*: Improving sensitivity to sub-ppt levels would require extensive re-engineering of the chemical recognition, a task more suited for a multi-year research project. In contrast, assay time is governed by measurable and adjustable physical parameters (pore radius, contact angle, film thickness), allowing quantitative modeling and realistic optimization using thermodynamics and transport principles learned in Course X.
2. *Greater real-world impact on usability*: While the current LOD already meets the need of preliminary field screen, the 30 minute assay time limits user adoption. Most commercial lateral flow devices deliver within 10-30 minutes. Reducing assay time to this benchmark would make the sensor competitive with existing LFA technologies and more attractive for portable, routine water testing.
3. *Direct relation to dimensionless transport analysis*: Time optimization leverages dimensionless modeling to link fluid flow, diffusion, and doping kinetics. These parameters can be experimentally tuned, modeled, and validated, providing a strong educational and engineering foundation for the project.
4. *Potential secondary benefits*: Reducing transport resistance and diffusion path lengths can also indirectly improve signal stability and reproducibility, as faster equilibrium minimizes hydration gradients and measurement drift. Thus, optimizing assay time contributes both to user practicality and sensor performance.

For these reasons, our design improvement effort will specifically target transport-limited performance. By modifying the F-PANI film structure and test-line geometry, we aim to reduce the total response from ~30 minutes closer to competitive times while maintaining ppt-level sensitivity and field-deployable simplicity.

9. Key Parameters

Table 3. *Parameters used for the design equations and calculations.*

Variable	Definition	Appears in	Range
L_1	Liquid travel distance	$t_{capillary}$	4 cm
L_2	F-PANI film thickness	t_{diff}, t_{ext}, Biot number, Da_{II}	7-10 µm, [6]
L_3	Characteristic pore length	Reynolds number	0.450 µm
ρ	density	Reynolds number	1 g/mL (water)
u	velocity	Reynolds number	$2x10^{-4}$ m/s
µ	Viscosity of liquid	$t_{capillary}$	$1 \cdot 10^{-3}$ Pa·s (water)
γ	Surface tension	$t_{capillary}$	72 mN/m [13]
r	Pore radius	$t_{capillary}$	0.225 µm [14] (range: 0.225-0.5 um)
θ	Contact angle of fluid	$t_{capillary}$	cos(147°) (range: 20-70°)
δ	Boundary layer of F-PANI film line	t_{ext}	27. 9 µm [6]
k_m	Mass transfer coefficient	t_{ext}, Biot number, Sherwood number	$1 \cdot 10^{-8}$ - $4 \cdot 10^{-5}$ m/s
D	Diffusivity of analyte	t_{diff}, Biot number, Da_{II}	10^{-12} - 10^{-10}
D_m	Diffusivity of analyte in water	t_{ext}	$5 \cdot 10^{-10}$

k_{on}	Association rate constant	t_{rxn}, Da_{II}	10^4 - 10^6
C_f	PFAS concentration in film	t_{rxn}, Da_{II}	10^{-10} M of PFOA EPA: 4 ppt or 10^{-12} M
k_{off}	Dissociation rate constant	t_{rxn}, Da_{II}	10^{-5} - 10^{-3}
l_m	NC membrane thickness	Tortuosity	200 µm [15]

10. Thermodynamic and Kinetic Constants

Table 4. *Applicable Thermodynamic and Kinetic Constants*

Thermodynamic or Kinetic Constant	**Calculated or Estimated number**
Reynolds number $Re = \frac{\rho u L_3}{\mu}$	$Re = \frac{1\times10^6 \cdot 2\times10^{-4} \cdot 0.45\times10^{-6}}{1} = 0.9 \times 10^{-4}$
Sherwood number $Sh = \frac{k_m L_2}{D}$	Reasonable assumption to be Sh = 2-20 due to laminar flow [16] Sh = 2
Damköhler number $Da_{II} = \frac{k_{on} C_f L_2^2}{D}$	$Da_{II} = \frac{10^5 \cdot 10^{-10} \cdot 0.45 \cdot 10^{-6}}{10^{-11}} = 0.45$
Biot number = $\frac{k_m L_2}{D}$	Assuming Sh = 2, $k_m = \frac{Sh \cdot D}{L_2} = \frac{2 \cdot 10^{-11}}{0.45 \cdot 10^{-6}} = 4.44 \times 10^{-5}$ Therefore, Bi = $\frac{4.44 \cdot 10^{-5} \cdot 0.45 \cdot 10^{-6}}{10^{-11}} = 2$
Peclet number $Pe = \frac{uL}{D}$	$Pe = \frac{2\times10^{-4} \cdot 0.45 \cdot 10^{-6}}{10^{-11}} = 9$

Tortuosity $\tau = \frac{L_3}{l_m}$	$\tau = \frac{0.45}{200} = 0.00225$

Based on the calculated dimensionless numbers, several transport characteristics can be inferred for the drive. The Reynolds number ($Re \sim 10^{-4}$) confirms that the flow within the lateral flow strip is laminar, as expected for capillary-driven systems [17, 18, 19]. The Sherwood number is assumed to be 2, which is reasonable under laminar flow conditions. The Damköhler number ($Da_{II} = 0.45$) indicates that the system operates in an intermediate regime, where it is neither fully reaction-limited (Da << 1) nor diffusion-limited (Da >>1). This suggests that both the surface reaction kinetics and mass transport processes contribute significantly to the overall performance and should be considered in the design equations. The Biot number ($Bi = 2$) further supports this conclusion, implying that internal and external mass-transfer resistances are comparable. The Peclet number ($Pe = 9$) suggest that both convection and diffusion play important roles in analyte transport across the test line. Finally, the tortuosity ($\tau = 0.00225$) indicates that fluid experiences minimal resistance and flow with little deviation from a straight path through the membrane. Overall, these dimensionless parameters confirm that the device operates under laminar, mixed transport conditions, where both reaction kinetics and diffusion jointly govern PFAS detection performance.

11. Design Equation

The total sensor response time (t_{total}) for PFAS detection using the F-PANI e-LFA is governed by the sequential transport and reaction processes occurring from the bulk liquid to the sensing polymer film. These include:

1. Capillary flow through the porous nitrocellulose membrane to deliver analyte to the F-PANI test line.
2. External mass transfer of PFAS molecules from the bulk liquid phase to the film surface.
3. Internal diffusion of PFAS through the F-PANI matrix to the protonation sites.
4. Reaction kinetics describing the protonation of PANI by the PFAS molecules.

Each process contributes to the characteristic time and the total response time is expressed as:

$$t_{total} = t_{capillary} + t_{external} + t_{internal} + t_{reaction}$$

The following assumptions were made in deriving the design equations:

- The liquid phase is incompressible and Newtonian.
- Flow within the porous membrane is laminar (Re<<1).
- The capillary channel has a uniform radius and constant contact angle.
- The contact angle $\theta < 90^{o}$, ensuring spontaneous imbibition.
- PFAS surface concentration is initially negligible ($C_s = 0$ at $t = 0$).
- Diffusion within the film and boundary layer is one-dimensional.
- Temperature, and hence properties (μ,γ,D), are constant.
- Convection within the F-PANI layer is neglected.
- PFAS–PANI protonation follows first-order Langmuir kinetics.
- The total response time is the sum of the sequential time constants.

<u>Capillary Flow</u>

Liquid transport through the NC membrane is governed by the Washburn equation, which can be seen in the following derivation, relating capillary pressure and viscous resistance:

1. Capillary pressure:

$$Pc = \frac{2\gamma \cos\theta}{r}$$

2. Gravitational pressure:

$$P_g = \rho g L(t)$$

$$\Delta P = P_c - P_g = \frac{2\gamma \cos\theta}{r} - \rho g L(t)$$

3. Poiseuille's law:

$$Q = \frac{\pi r^4 \Delta P}{8\mu L}$$

4. Solving differential equation:

$$Q = \frac{dV}{dt} = \pi r^2 \frac{dL}{dt}$$

$$\pi r^2 \frac{dL}{dt} = \frac{\pi r^4 \Delta P}{8\mu L} = \frac{\pi r^4}{8\mu L}\left(\frac{2\gamma \cos\theta}{r} - \rho g h(t)\right)$$

$$L = \sqrt{\frac{(r\gamma cos\theta t)}{2\eta}}$$

5. Simplification and assumptions for final rearrangement:

$$t_{wick} = \frac{2\mu L_1^{\ 2}}{\gamma r cos\theta}$$

where μ is the fluid viscosity, L_1 is the travel length of the liquid front, γ is the surface tension, r is the average pore radius, and θ is the contact angle. This represents the time required for the wetting front to reach the F-PANI test line.

<u>External Mass Transfer</u>

Once the test line is wetted, the rate of mass transfer of PFAS molecules diffuse from the bulk liquid to the F-PANI surface can be described by the mass transfer equation:

$$J = k_m(C_b - C_s)$$

where J is the mass flux, k_m is the mass transfer coefficient, C_b is bulk concentration and C_s is surface concentration. For the initial stage of the process, we assume $C_s \approx 0$ (negligible surface concentration at $t = 0$), so:

$$J = k_m C_b$$

The amount of PFAS that needs to be transferred to the surface per unit area is related to the effective thickness δ of the diffusion boundary layer and the bulk concentration C_b. The mass per unit area to be transferred is approximately $C_b \times \delta$. The characteristic time t_{ext} is the time required to establish a significant fraction of the equilibrium surface concentration, which can be approximated by the time to transfer this mass over the area:

$$t_{ext} = \frac{mass\ per\ unit\ area}{mass\ flux\ per\ unit\ area} = \frac{C_b \times \delta}{k_m \times C_b} = \frac{\delta}{k_m}$$

The external mass transfer coefficient (k_m) can be estimated using a mass transfer correlation for flow through a porous medium, such as the Sherwood number (Sh) relation:

$$Sh = \frac{k_m L_2}{D_m}$$

For low Re, $Sh \approx 2$, so:

$$k_m = \frac{2D_m}{L_2}$$

So final equation for t_{ext} becomes:

$$t_{ext} = \frac{L_2 \times \delta}{2D_m}$$

<u>Internal Mass Transfer</u>

After reaching the F-PANI, PFAS diffuses through the polymer layer of thickness L_2. For one-dimensional diffusion described by Fick's second law, the characteristic diffusion time for internal mass transfer can be derived by:

1. Governing Equation: Fick's Second Law:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

2. Boundary conditions:

$$C(0,\ t) = C_s \text{ at } x = 0$$

$$C(x,\ 0) = 0 \text{ at } t = 0$$

$$\frac{\partial C}{\partial x} = 0 \text{ at } x = \delta$$

3. Dimensionless Number Analysis: To derive the characteristic time t_{int}, balance the dimensions of the terms in Fick's second law.

   Left side ($\frac{\partial C}{\partial t}$): Dimensions of concentration per time, $[C]\ /\ [T]$.

   Right side ($D\frac{\partial^2 C}{\partial x^2}$): D has dimensions $[L]^2/[T]$, and $\frac{\partial^2 C}{\partial x^2}$ has $[C]\ /\ [L]^2$,so overall $[C]/[T]$.

   Approximate the derivatives with finite differences over characteristic scales: Let ΔC be the concentration change (e.g., from 0 to C_s). The time derivative scales as $\Delta C/\ t_{ext}$, and the spatial derivative as $\Delta C/L^2$.

   Substituting:

$$\Delta C/\ t_{int} \sim\ D\ *\ \Delta C/\delta^2$$

   Cancel ΔC:

$$1/t_{int} \sim\ D\ *\ 1/\delta^2$$

$$t_{int} = \frac{\delta^2}{D}$$

Reaction Kinetics

Inside the film, PFAS molecules protonate the emeraldine base of the PANI following first-order Langmuir kinetics:

$$\frac{dC}{dt} = k_{on}C_f(1 - \theta) - k_{off}\theta$$

The characteristic time is:

$$t_{rxn} = \frac{1}{k_{on}C_f + k_{off}}$$

where k_{on} and k_{off} are the association and dissociation rate constants respectively, and C_f is the PFAS concentration at the film interface.

Final Design Equation

Combining all contributing terms yields the complete design equation for the sensor response time:

$$t_{total} = \frac{2\mu L_1^2}{\gamma r cos\theta} + \frac{L_2 \times \delta}{2D_m} + \frac{\delta^2}{D} + \frac{1}{k_{on}C_f + k_{off}}$$

This expression captures all the relevant physical and kinetic contributions to the overall equilibration time of the F-PANI e-LFA sensor.

12. Results and Discussion

Washburn Time Plots

The objective is to minimize the total time (t_{total}) by modeling t_{wash}, which is governed by contact angle, flow distance, and pore radius. The contact angle (θ) of ranging from 20° to 70°, pore radius (r) of 0.225 um to 0.5 um, and the strip length (L) range of 2 to 4 cm were chosen for modeling. Park et al. indicated that lower contact angle results in faster capillary flow. Therefore, the contact angle range of 20° to 70° was selected, avoiding the extreme cases at cos(0°) and cos(90°). The pore radius range was selected based on the range produced by commercial nitrocellulose membrane manufacturers [20]. The minimum strip length is constrained by 0.7 cm thickness of the test line. Park et al. used a strip length of 4 cm. Examination of the t_{wash} equation indicates that decreasing the strip length L reduces the time.

However, additional space before and after test line is required for proper flow, and therefore we chose *L* ranging from 2 to 4 cm. Three cases were modeled depicting one "constant" variable and two varying variables.

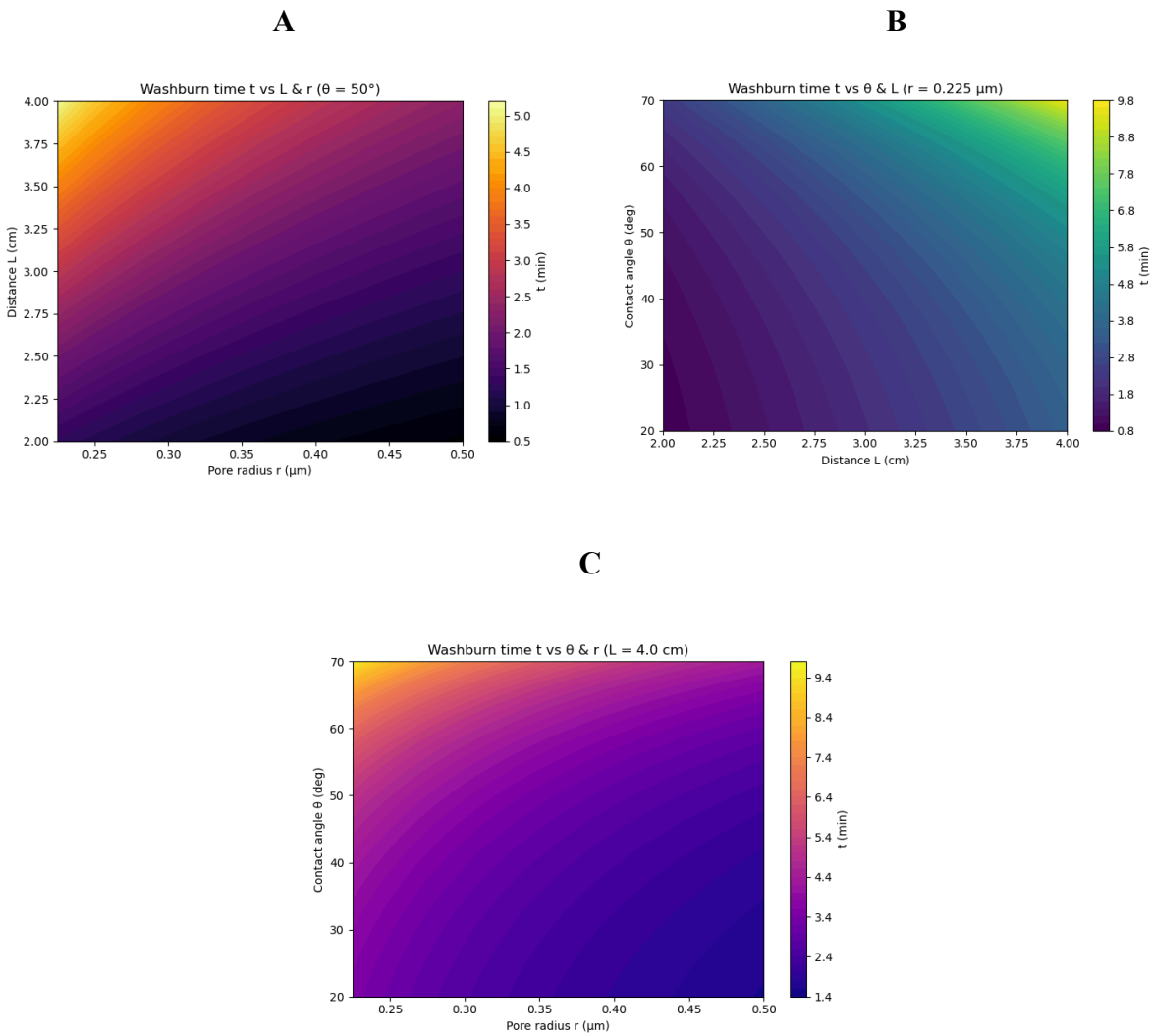

Figure 8. *Washburn Time Plots:* ***(A)*** *θ fixed at 50°, showing the effect of varying r and L,* ***(B)*** *r fixed at 0.225 μm, showing the effect of varying θ and L ,* ***(C)*** *L fixed at 4 cm, showing the effect of varying θ and r*

When the contact angle is fixed to 50°, a smaller distance of 2 cm and large pore radius of 0.50 um optimized the time to 0.5 min (Figure 8A). A fixed radius of 0.225 um shows a lower distance of 2 cm and low contact angle of 20° to optimize the time to 0.8 min (Figure 8B). When the distance is fixed to 4 cm, a high pore radius of 0.50 um and low contact angle of 20° provides

a time of 1.4 min (Figure 8C). Given Graph 1 produces the lowest time of 0.5 min, the optimal conditions for design are a contact angle = 50°, pore radius = 0.50 µm, and distance = 2.00 cm.

Analyte Concentration Plot

Another factor that contributes significantly to the total time is the reaction time, t_{rxn}, which depends on the association rate constant (k_{on}), dissociation rate constant (k_{off}), and analyte concentration (C_f). We set the $k_{on} = 10^6$ and $k_{off} = 10^{-3}$. These parameter values were selected to closely approximate the experimental data reported by Park et al. In addition, the analyte concentration range was based on Park et al. as well, where the concentration was varied between 10^{-10} - 10^{-2} M.

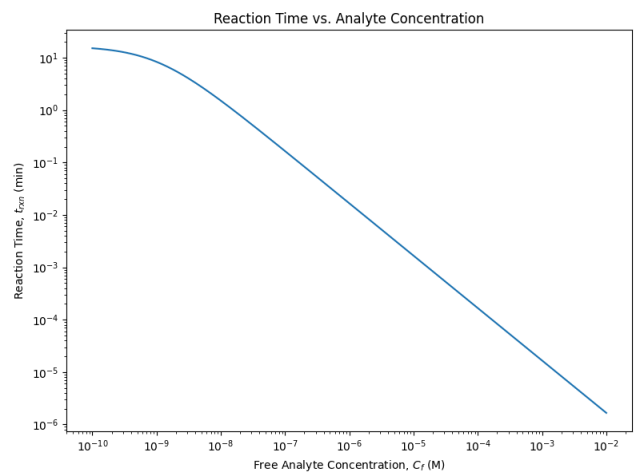

Figure 9. *Reaction time change in terms of analyte concentration*

The reaction time and analyte concentration have an inverse relationship (Figure 9). According to Park et al. , 10^{-9} M corresponds with the 400 ppt limit of detection for PFOA. Therefore, the corresponding time at 10^{-9} M was determined to be the optimal reaction time.

Total time plots

To estimate the total time, individual time contributions were calculated based on the parameters reported in Table 3. The test line thickness $L_2 = 0.7\ cm$, boundary layer thickness $\delta = 27.9\ \mu m$, diffusion coefficients of $D_m = 5 \times 10^{-10}$ in the membrane and $D = 10^{-11}$ for the analyte in the solution used for the calculations. The external diffusion time was calculated as

$t_{ext} = 3.26\ min$ and internal diffusion time $t_{int} = 1.3\ min$. The reaction time was estimated to be $t_{rxn} = 8.33\ min$ for analyte concentration of $C_f = 10^{-9} M$. This gives us $t_{total} = t(L, \theta) + 12.89\ min$. Detailed calculations can be found in Appendix B.

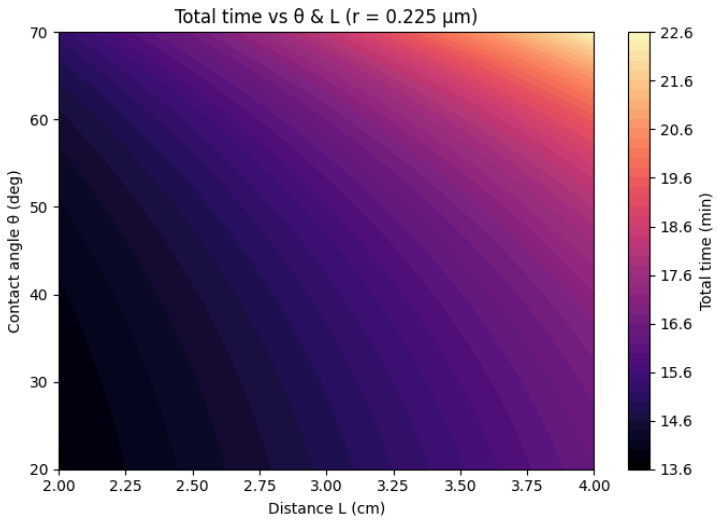

Figure 10. *Effect of varying L and θ on total time*

Figure 10 revealed that lowering contact angle and distance reduces total time, with the lowest time of 13.6 min occurring in the region with the smallest distance and the lowest contact angle. In addition, the gradient of the plot indicated that lowering distance has a stronger influence on total time than contact angle. This suggests that minimizing transport distance is a more effective strategy for improving performance. However, changes in contact angle still contribute meaningfully and may interact with distance to further optimize overall time.

13. Tradeoffs and Discussion

While the reduction of time is advantageous for consumers, the device design alterations come with several constraints. By decreasing the time necessary for the resistivity signal to stabilize, there is a decreased time for interactions between the PFAS aqueous solution and the F-PANI test line. This could lead to a weaker resistivity signal due to a reduced time for PANI-EB to convert to PANI-ES. In addition, a lower signal could lead to lower detection rates of PFAS. Currently, the device can detect PFOA at 400 ppt. However, with EPA limits as low as 4 ppt, a weaker signal could lead to false negative results and reduced accuracy in the device's ability to detect PFAS. Therefore, future work includes the reduction of the limit of detection.

One possible solution would be to increase the surface area of the nanofibers of F-PANI in order to increase the interaction sites.

A second constraint is the ease of manufacturing the new design. A change in device parameters can lead to additional manufacturing steps, which affects cost and time. One varied parameter is the distance the liquid travels. The optimized time includes a decrease in distance liquid travels to 2 cm. NC membranes are typically manufactured with a length of 7-9 cm [14]. Therefore, an additional step to fit the membrane to the optimized size would need to be added to the design process. However, the decrease in membrane length may reduce the cost associated with the production as multiple optimized NC membranes could be created from a larger NC membrane.

14. Contributions

G. Dovranova: Sections 9, 10, 11, 12, 16
J. Chen: Sections 4, 5, 9, 10, 11, 15
M. Wang: Sections 1, 2, 3, 8, 9, 10, 11, 16
S. Thompson: Sections 6, 7, 9, 10, 11, 12, 13, 16
All group members contributed to making the presentation and presenting.

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16. Appendix

A. Modeling Porous Membranes

In addition to design equations for the design of the assay, the tortuosity can also be calculated to best describe the fluid flow through the membrane.

1. Hagen - Pouseille Equation

$$v = \frac{D^2}{32\mu L}(Po - Pl)$$

2. Extent equation

$$\epsilon = \frac{n\pi D^2}{4}$$

3. Straight pores of uniform diameter

$$N = \frac{n\pi D^2 \rho}{128\mu l}(Po - Pl)$$

4. Compensation for pore diameter variability

$$L = lm\tau$$

$$d = \frac{4\epsilon}{a(1-\epsilon)}$$

$$N = \frac{\rho\epsilon^3(Po - Pl)}{2(1-\epsilon)^2\tau a^2\mu l}$$

B. Time Calculations

$$t_{ext} = \frac{L_2 \times \delta}{2D_m} = \frac{(0.007)\times(27.9\times10^{-6})}{2\times5\times10^{-10}} = 195.3\ s = 3.26\ min$$

$$t_{int} = \frac{\delta^2}{D} = \frac{(27.9\times10^{-6})^2}{10^{-11}} = 78\ s = 1.3\ min$$

$$t_{rxn} = \frac{1}{k_{on}C_f + k_{off}} = \frac{1}{(1\times10^6)\times(1\times10^{-9})+(10^{-3})} = 8.33\ min$$

$$t_{total} = t_{capillary} + t_{external} + t_{internal} + t_{reaction} = t(r,\ \theta) + 3.26 + 1.3 + 8.33 = t(r,\ \theta) + 12.89$$